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2nd year /Electronic I
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The Field-Effect Transistor

Constant-Current Biasing:

Example 3.3: Determine the currents and voltages in a MOSFET constant-current source. For the circuit shown in Figure 3.3, the transistor parameters are: K_{n1} = 0.2 mA/V^2, $K_{n2} = K_{n3} = K_{n4}$ = 0.1 mA/V^2, and $V_{TN1} = V_{TN2} = V_{TN3} = V_{TN4}$ = 1_V.

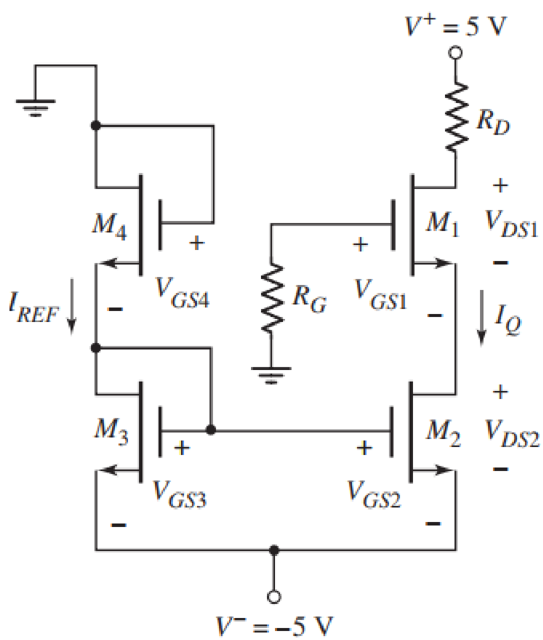

Figure 3.3 Implementation of a MOSFET constant-current source

Solution: **Step 1 V_{GS3}**

Since the reference current is the same in transistors M3 and M4, then

$K_{n3}(V_{GS3} - V_{TN3})^2 = K_{n4}(V_{GS4} - V_{TN4})^2$

$V_{GS4} + V_{GS3} = -V-$

This yield to V_{GS3}= 2.5 V which is equal to V_{GS4}

Step 2 I_Q

Since M_3 and M_4 are identical transistors, V_{GS3} should be one-half of the bias voltage. The bias current I_Q is then

$I_Q = (0.1) \cdot (2.5 - 1)^2 = 0.225$ mA

Step 3 V_{GS1}

The gate-to-source voltage on M1 is found from

$I_Q = K_{n1}(V_{GS1} - V_{TN1})^2$

$0.225 = (0.2) \cdot (V_{GS1} - 1)^2$

which yields

$V_{GS1} = 2.06$ V

Step 4 V_{DS2}

The drain-to-source voltage on M2 is

$V_{DS2} = (-V -) - V_{GS1} = 5 - 2.06 = 2.94$ V

Since $V_{DS2} = 2.94$ V $> V_{DS}$ (sat) $= V_{GS2} - V_{TN2} = 2.5 - 1 = 1.5$ V, M_2 is biased in the saturation region.

Multistage MOSFET Circuits

Cascade Configuration

Example 3.3: Design the biasing of a multistage MOSFET circuit to meet specific requirements. Consider the circuit shown in Figure 3.6 with transistor parameters K_{n1} = 500 μA/V^2, K_{n2} = 200 μA/V^2, $V_{TN1} = V_{TN2}$ = 1.2 V, and λ1 = λ2 = 0.

Design the circuit such that <u>I_{DQ1} = 0.2 mA, I_{DQ2} = 0.5 mA</u>, <u>$V_{DSQ1} = V_{DSQ2}$ = 6V</u>, and <u>Ri = 100 kΩ</u>. Let R_{Si} = 4 kΩ.

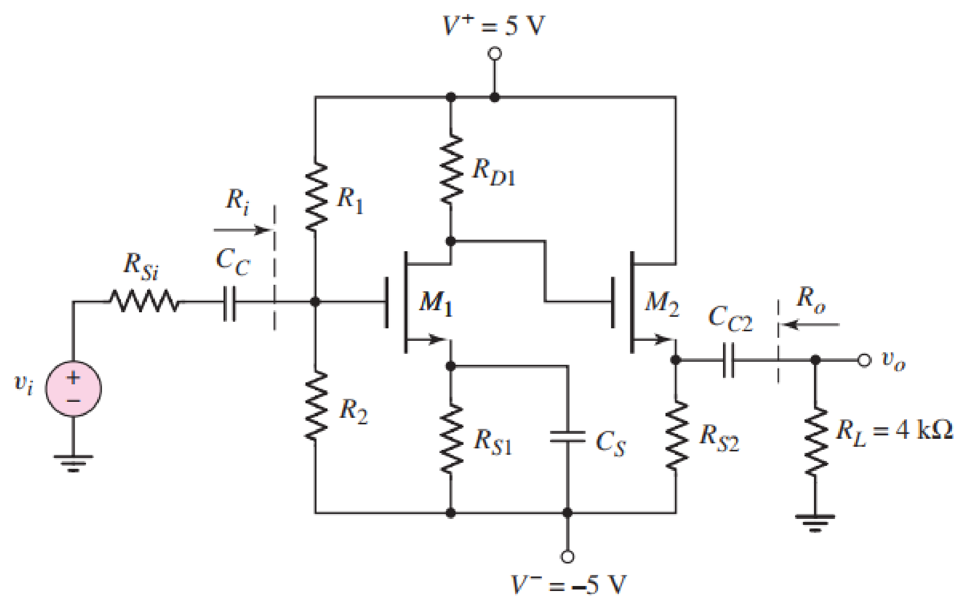

Figure 3.6 Common-source amplifier in cascade with source follower

Solution:

<u>Step 1 R_{S2}</u>

For output transistor M_2, we have

$V_{DSQ2} = 5 - (-5) - I_{DQ2} R_{S2}$

or $6 = 10 - (0.5) R_{S2}$

which yields R_{S2} = 8 kΩ.

Step 2 R_{D1}

Assuming transistors are biased in the saturation

region and the value of K_{GS2}

$I_{DQ2} = K_{n2}(K_{GS2} - V_{TN2})^2$

or $0.5 = 0.2(V_{GS2} - 1.2)^2$

which yields $V_{GS2} = 2.78$ V

Since $V_{DSQ2} = 6$V, the source voltage of M_2 is $V_{S2} = -1$ V. With $V_{GS2} =$ 2.78 V, the gate voltage on M_2 must be

$V_{G2} = -1 + 2.78 = 1.78$ V

$R_{D1} = [5 - 1.78]/ 0.2= 16.1$ kΩ

Step 3 R_{S1} is then

For $V_{DSQ1} = 6$ V, the source voltage of M_1 is

$V_{S1} = 1.78 - 6 = -4.22$ V

$R_{S1} = [-4.22 - (-5)]/ 0.2 = 3.9$ kΩ.

Step 4 R_1 and R_2

For transistor M_1, we have

$I_{DQ1} = K_{n1} (V_{GS1} - V_{TN1})^2$

or $0.2 = 0.50(V_{GS1} - 1.2)^2$

which yields $V_{GS1} = 1.83$ V

To find R_1 and R_2, we can write

$$V_{GS1} = \left(\frac{R_2}{R_1 + R_2}\right)(10) - I_{DQ1}R_{S1}$$

$$\frac{R_2}{R_1 + R_2} = \frac{1}{R_1} \cdot \left(\frac{R_1 R_2}{R_1 + R_2}\right) = \frac{1}{R_1} \cdot R_i$$

then, since the input resistance is specified to be 100 k, we have

$$1.83 = \frac{1}{R_1}(100)(10) - (0.2)(3.9)$$

which yields R_1 = 383 kΩ. From R_i = 100 kΩ, we find that R_2 = 135 kΩ.

Example 3.4: Design the biasing of the cascode circuit to meet specific requirements. For the circuit shown in Figure 3.7, the transistor parameters are: $V_{TN1} = V_{TN2}$ = 1.2 V, $K_{n1} = K_{n2}$ = 0.8 mA/V^2, and $\lambda 1 = \lambda 2 = 0$. Let $R_1 + R_2 + R_3$ = 300 kΩ and R_S = 10kΩ. Design the circuit such that I_{DQ} = 0.4 mA and $V_{DSQ1} = V_{DSQ2}$ = 2.5 V.

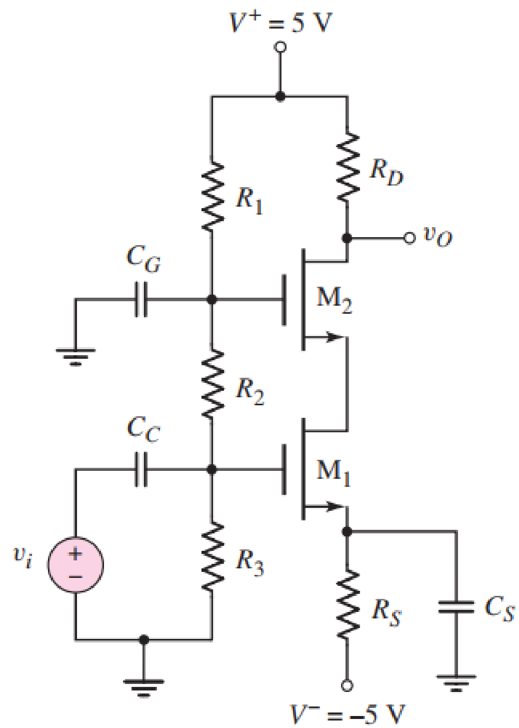

Figure 3.7 NMOS cascode circuit

Solution:

Step 1 R_3

The dc voltage at the source of M_1 is

$V_{S1} = I_{DQ} R_S - 5 = (0.4)(10) - 5 = -1V$

Since M_1 and M_2 are identical transistors, and since the same current exists in the two transistors, the gate-to-source voltage is the same for both devices. We have

$I_D = Kn (V_{GS} - V_{TN})^2$

or

$0.4 = 0.8(V_{GS} - 1.2)^2$

which yields

$V_{GS} = 1.907$ V

Then,

$$V_{G1} = \left(\frac{R_3}{R_1 + R_2 + R_3}\right)(5) = V_{GS} + V_{S1}$$

$$\left(\frac{R_3}{300}\right)(5) = 1.907 - 1 = 0.907$$

which yields

$R_3 = 54.4$ kΩ.

<u>Step 2 R_2</u>

The voltage at the source of M2 is

$V_{S2} = V_{DSQ1} + V_{S1} = 2.5 - 1 = 1.5$ V

Then,

$$V_{G2} = \left(\frac{R_2 + R_3}{R_1 + R_2 + R_3}\right)(5) = V_{GS} + V_{S2}$$

$$\left(\frac{R_2 + R_3}{300}\right)(5) = 1.907 + 1.5 = 3.407 \text{ V}$$

which yields

$R_2 + R_3 = 204.4\ \text{k}\Omega$

and

$R_2 = 150\ \text{k}\Omega$

Step 3 R_1

Therefore

$R_1 = 95.6\ \text{k}\Omega$

Step 4 R_D

The voltage at the drain of M_2 is

$V_{D2} = V_{DSQ2} + V_{S2} = 2.5 + 1.5 = 4$ V

The drain resistor is therefore

$$R_D = \frac{5 - V_{D2}}{I_{DQ}} = \frac{5-4}{0.4} = 2.5\ \text{k}\Omega$$

Note: Since $V_{DS} = 2.5$ V > $V_{GS} - V_{TN} = 1.91 - 1.2 = 0.71$ V, each transistor is biased in the saturation region.